Michalis Chadolias

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Experience

Predoctoral Researcher — Feb 2026 – Present
Instituto de Física Corpuscular (IFIC) - VEGA Group — *Valencia, Spain – Full-time*

- Planned contribution to the Data/MC comparison efforts for the ORCA23/24 data, focusing on event classification and spark rejection for background reduction through BDTs
- Position funded through the FPI fellowship from the Spanish Ministry of Science and Innovation.

Erlangen Centre for Astroparticle Physics (ECAP) — Erlangen, Germany
Master Thesis Student — *Aug 2023 – Oct 2024, Full-time*

- Engineered a scalable, automated data pipeline for preprocessing 500k+ simulation files, improving speed throughput by 20% and reducing storage by 35%.
- Developed automated, parallel workflows (bash + SLURM) for large-scale parameter sweeps on HPC clusters (NHR@FAU, CC@Lyon), enabling reproducible simulation and analysis runs.
- Executed large-scale parameter sweeps across 200+ model configurations using reproducible workflows, evaluating model performance and systematic uncertainties.

Erlangen Centre for Astroparticle Physics (ECAP) — Erlangen, Germany
Research Assistant (HiWi) — *Nov 2022 – Apr 2023, Part-time*

- Engineered a robust Monte Carlo production pipeline using Snakemake, resulting in a reduction of processing errors by 30% and establishing a more reliable framework for complex data simulations in research projects. .
- Diagnosed and optimised distributed workflow executions, reducing job failures and improving reproducibility on HPC environments.
- Optimised SLURM resource profiles, achieving a 25% CPU workload reduction through improved parallelisation and job scheduling.

Projects

Detect Pulsar API | Python, FastAPI, Docker, XGBoost, SHAP — mchadolias/detect-pulsar-api

- Built an end-to-end supervised learning pipeline for blazar source classification, including data preprocessing, feature engineering, and model benchmarking with XGBoost.
- Developed a production-ready FastAPI prediction service and containerised the entire application with Docker for deployment.
- Applied SHAP-based explainability to interpret model behaviour, evaluate feature contributions, and ensure model robustness.

MiniBooNE Event Classifier | Python, scikit-learn, Optuna, MLflow — mchadolias/miniboone-classifier

- Applied supervised learning methods for binary classification of particle events, including preprocessing, feature scaling, and algorithm comparison.
- Performed hyperparameter optimisation using ASHA via Optuna to efficiently explore the model search space and improve predictive performance.
- Tracked experiments and model artefacts with MLflow to enable reproducible workflows and systematic evaluation.

Skills

Programming: Python, C/C++, Bash, SQL, NumPy, Pandas, Matplotlib, Seaborn, Plotly
ML & Data Tools: scikit-learn, XGBoost, Tensorflow, PyTorch, SHAP, Optuna, MLflow, Snakemake
Deployment & DevOps: FastAPI, Flask, Docker, Apptainer, CI/CD (GitHub Actions/Gitlab), SLURM
Foreign Languages: Greek (Native), English (C2), German (B2), Italian (A1)

Education

University of Valencia (UV) — Valencia, Spain
Ph.D in Neutrino Physics — *Feb 2026 – Present*

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) — Erlangen, Germany
M.Sc. in Physics, Astroparticle Specialisation — Grade: 90/100 — *Apr 2022 – Feb 2025*

Aristotle University of Thessaloniki (AUTh) — Thessaloniki, Greece
B.Sc. in Physics — Grade: 7.84 / 10.00 — *Oct 2017 – Feb 2022*